

pamica: GPU-accelerated Adaptive Mixture Independent Component Analysis in Python with Fortran parity

Seyed Yahya Shirazi¹, Arnaud Delorme^{1,2}, and Scott Makeig¹

¹ Swartz Center for Computational Neuroscience, Institute for Neural Computation, University of California San Diego, USA ² Centre de Recherche Cerveau et Cognition (CerCo), CNRS, University of Toulouse, France ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Open Journals

Reviewers:

- @openjournals

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

Summary

Independent Component Analysis (ICA) is a standard method for separating electroencephalography (EEG) and electromyography (EMG) recordings into maximally independent sources, which isolates brain, muscle, and artifact activity for downstream analysis. Adaptive Mixture ICA (AMICA) generalizes single-model ICA to a mixture of ICA models with adaptive source densities, and produces the most dipolar (and thus most physiologically interpretable) component decompositions of EEG among the widely used algorithms benchmarked by Delorme et al. (2012). Its reference implementation is a Fortran program parallelized with the Message Passing Interface (MPI) and distributed as a compiled binary driven from MATLAB/EEGLAB, which is difficult to install, runs only on the CPU, and is not usable from a Python scientific workflow.

pamica is a Python implementation of AMICA that reproduces the reference Fortran results within numerical tolerance while running on the CPU, NVIDIA GPUs (CUDA), and Apple GPUs (Apple's MLX array framework (Hannun et al., 2023)). It is a complete NumPy/PyTorch reimplementation, not a wrapper around the Fortran binary, built on PyTorch (Paszke et al., 2019), NumPy (Harris et al., 2020), and SciPy (Virtanen et al., 2020) (Python \$ \$3.12, PyTorch \$ \$2.12; tested on Apple Silicon/ARM64 and x86_64/CUDA Linux), and exposes a scikit-learn-style estimator under a BSD-3-Clause license. In double precision it reproduces the reference score-function algebra to machine precision; it also runs in single precision (float32), which the CPU-only binary does not offer and which is numerically faithful (agreeing with double precision to four to five significant digits) yet required to use Apple GPUs, which have no float64 and host the fastest backend (MLX). pamica writes output in the binary format that EEGLAB's AMICA loader reads: a single-model output file is byte-identical in layout to a native AMICA file and needs no manual re-interpretation, and multi-model output round-trips through the same loader. Correctness is defined as parity with the Fortran reference for the single-model case and, because multi-model AMICA is not partition-identifiable, as a similar distribution of solutions for the multi-model case; both are validated on real EEG against the reference binary. A minimal example is a scikit-learn-style call, `AMICA(n_models=1, n_mix=3).fit(X)` on a (channels, samples) array (full workflow in the README). The software is at <https://github.com/scen/pAMICA> (archived at doi:10.5281/zenodo.21312148).

Statement of need

AMICA yields components that are well suited to equivalent-dipole source localization and to automated classification of independent components (Pion-Tonachini et al., 2019), and it separates EEG more effectively than most alternatives (Delorme et al., 2012; Palmer et al.,

2012). The reference Fortran implementation shares these limitations and is invoked as an external process from MATLAB rather than called from Python. As neuroimaging analysis has moved toward Python, for example MNE-Python (Gramfort et al., 2013), an AMICA that runs natively in Python and on a GPU, and that is validated to reproduce the Fortran reference numerically, is needed for modern pipelines and for studying the algorithm in an open codebase.

General-purpose Python ICA implementations do not fill this gap. scikit-learn and MNE-Python provide FastICA (Hyvärinen & Oja, 2000) and Infomax (Bell & Sejnowski, 1995; Lee et al., 1999), and Picard (Ablin et al., 2018) provides a faster maximum-likelihood ICA, but none implement AMICA's mixture of models, adaptive generalized-Gaussian source densities, or Newton updates, and so they do not reproduce AMICA decompositions. pamica targets researchers who need AMICA: EEG/EMG analysts who want AMICA-quality decompositions inside a Python pipeline, users of GPU hardware who want faster runs than the CPU-only binary, and methodologists who need a transparent reference implementation to inspect and build on.

Implementation and validation

pamica provides a natural-gradient (Amari, 1998) expectation-maximization backend that ports the full AMICA algorithm: exact-EM mixture updates, a positive-definite Newton step (Palmer et al., 2008), symmetric zero-phase-component-analysis (ZCA) sphering, the five source-density families of the reference (generalized Gaussian, Gaussian, logistic, sub-Gaussian, and the extended-Infomax kurtosis switcher), a mixture of ICA models, and component sharing across models.

pamica's conformity with the reference binary is measured with two complementary metrics: Hungarian-matched component correlation, and the Amari distance (Amari et al., 1996), a standard unmixing-matrix comparison metric that needs no assignment step since it is permutation- and scale-invariant by construction. Both implementations were run for AMICA's usual 2000 iterations with Newton off (`do_newton=0`) and otherwise-default parameters (two separate configuration files drive them, JSON for pamica and Fortran's own text format, with a transcribed subset of settings; a shared-format reader is planned). Newton is disabled here to isolate the algorithm from initialization: with Newton on, independent random seeds (the backends share no random state) can settle a few under-determined components in different but equally likely optima, whereas from a matched initialization the two agree (documentation). The single-model headline (Table 1) uses an external recording (OpenNeuro ds002718, 70 channels, $k \approx 153$), well past the ~ 60 threshold where cross-backend agreement plateaus; the bundled 32-channel sample below ($k \approx 30$) gives a consistent Amari distance. Score functions and per-block sufficient statistics are exact to floating-point resolution against the literal Fortran expressions on the bundled sample. A mixture of ICA models is not partition-identifiable, so exact partition parity is the wrong bar for the multi-model case; it is instead assessed by whether the two implementations sample a similar distribution of solutions, across ensembles of 20 bundled-sample runs each (Figure 1). A run-level permutation test (the 40 runs permuted as intact units) finds no evidence that cross-implementation agreement is worse than Fortran's own run-to-run agreement, so single-run values reflect intrinsic estimator spread rather than a shortfall. The multi-model log-likelihood distributions still differ slightly at a matched 100-iteration budget (pamica reaches Fortran's mean with about twice as many iterations), so full-likelihood similarity is not yet claimed.

Table 1: Single-model parity (external ds002718, $k \approx 153$; Amari and score-function checks on the bundled sample) and multi-model distributional similarity (bundled sample) of pamica with the Fortran reference. All values are means (sd shown where relevant) over matched components (single-model) or 190 within- and 400 cross-implementation pairs (multi-model). Full methodology and reproduction steps are in the documentation.

Regime	Metric	Result (mean; correlation / Amari distance)
Single-model	Log-likelihood gap to Fortran (mean per-sample-channel LL, -3.6993 , $k \approx 153$)	within ~ 0.0005
Single-model	Conformity with Fortran	0.998 ($k \approx 153$) / 0.006 (bundled, $k \approx 30$)
Single-model	Score functions (non-default families) and sufficient statistics	exact to floating-point resolution ($\sim 10^{-15}$)
Multi-model	Single-run magnitude (pamica-Fortran; Fortran-Fortran)	0.65 ; 0.64 (sd 0.05) / 0.163 ; 0.174 (sd 0.02)
Multi-model	Ensemble similarity, 20 runs each: mean difference, between — within-Fortran (permutation p)	$+0.011$ ($p = 0.96$) / -0.011 ($p > 0.999$)

Multi-model AMICA (n models=2): pyAMICA vs Fortran ensembles, real sample EEG

Dashed vertical lines mark each distribution's mean.

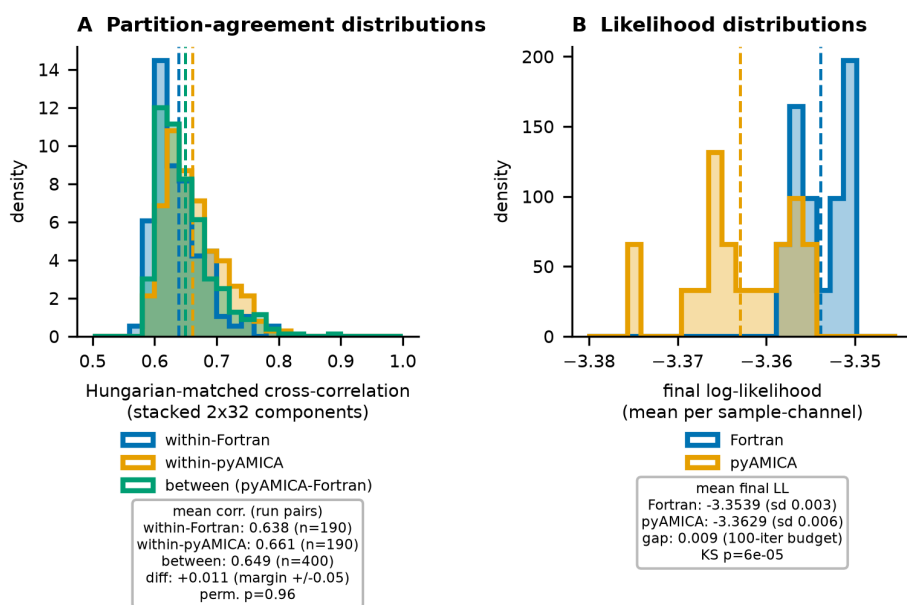


Figure 1: Multi-model solution-ensemble partition-correlation distributions (panel A) and log-likelihood distributions (panel B) for 20 pamica and 20 Fortran fits of the sample EEG; dashed lines mark each distribution's mean. The within-Fortran, within-pamica, and between-implementation correlation distributions overlap, so the single-run correlation reflects the estimator's intrinsic run-to-run spread rather than a gap to the reference. Panel B's apparent separation is a 0.009 log-likelihood gap on a ~ 0.035 axis.

86 All backends converge to the same single-model log-likelihood on real EEG (maximum pairwise
87 difference ~ 0.003). On Apple Silicon the MLX backend is the fastest option and is flat with
88 channel count (15-25 ms per iteration from 16 to 70 channels; see the documentation), about
89 eight times faster than double-precision multithreaded CPU; PyTorch-MPS is not a win (at or
90 worse than the CPU). On NVIDIA hardware double-precision CUDA is the reproducible path
91 and is overhead-bound at EEG scale, so single precision gives it little additional speedup (Table
92 2). Native Fortran itself scales with CPU cores, unlike the CPU backends above: with enough
93 cores pinned it beats the CUDA GPU on the workstation (Table 2), though only on a larger,
94 hotter host; it does not catch Apple's MLX on laptop hardware. A data-size sweep further
95 shows cross-backend component agreement rising with frames per channel and plateauing
96 near 0.98 once the decomposition is well-determined, where two independent double-precision
97 implementations (native Fortran and PyTorch-CUDA) agree at a mean of 0.995; single-precision
98 runs agree with double precision to four to five significant digits, so double precision remains
99 the default for parity.

Table 2: Single-model throughput on real 70-channel EEG ($n_{\text{mix}}=3$, $\text{pdftype}=0$, $\text{block_size}=512$; warm, minimum of repeated runs). CPU, MPS, and MLX on Apple Silicon; CUDA on a separate NVIDIA RTX 4090; CUDA float32 is comparable (~ 36 ms). The two native-Fortran rows are from a separate core-count sweep (documentation) at each backend's plateau; the other CPU rows use platform-default threads, so they are not core-matched to Fortran. Unlike the correctness comparison, this benchmark uses external data (OpenNeuro ds002718, one subject so far) and specific GPU hardware.

Backend (device)	Precision	ms / iteration
MLX (Apple GPU)	float32	25
CUDA (NVIDIA RTX 4090)	float64	39
Native Fortran (Intel Core i9-13900K, 24 cores)	float64	30
PyTorch CPU (Apple Silicon)	float64	193
Native Fortran (Apple Silicon, 8 cores)	float64	70
PyTorch MPS (Apple GPU)	float32	255
NumPy (reference, Apple Silicon)	float64	622

100 The correctness harness compares *pamica* against Fortran with two metrics, Hungarian-matched
101 component correlation and Amari distance, and never uses synthetic data; the multi-model and
102 score-function checks need no external download (bundled sample only). The full performance
103 tables, per-run Amari-distance detail, data-size sweep, and step-by-step reproduction commands
104 are in the documentation (<https://eeglab.org/pAMICA/guides/validation/>).

105 State of the field

106 *pamica* complements rather than replaces EEGLAB (Delorme & Makeig, 2004) and its Fortran
107 AMICA plugin: it reads and writes the same output format, so results move between the two,
108 adds GPU support and a Python API, and can run the reference Fortran itself from Python
109 through a bundled dependency-free native build (no MKL or MPI runtime). Two other Python
110 AMICA reimplementations have appeared (Esmaeili, 2025; Herforth, 2026), both of which
111 provide MNE-Python-compatible objects; *pamica* leads with a scikit-learn-style array API and
112 byte-identical EEGLAB I/O, and also ships an MNE-Python wrapper (an optional extra).
113 What distinguishes *pamica* is the rigor and scope of its Fortran-parity validation, beyond what
114 either alternative publishes: source-density score functions exact to floating-point resolution
115 ($\sim 10^{-15}$), single-model component correlation and Amari distance against the reference, and
116 a distributional-similarity framework for the non-identifiable multi-model case, together with
117 byte-identical EEGLAB output and an MLX backend for Apple GPUs.

Acknowledgements

We thank Jason Palmer and Ken Kreutz-Delgado, co-developers of AMICA, for the reference implementation, and the EEGLAB community for the tools and sample data used to validate this work. Two of the authors are original developers of the methods *pamica* builds on: S.M. co-developed the AMICA algorithm (Palmer et al., cited above) and A.D. is a lead developer of EEGLAB (Delorme & Makeig, 2004). This work was supported by The Swartz Foundation (Old Field, NY) to the Swartz Center for Computational Neuroscience and by National Institutes of Health grant R01-NS047293 (to A.D. and S.M.).

References

- Ablin, P., Cardoso, J.-F., & Gramfort, A. (2018). Faster independent component analysis by preconditioning with hessian approximations. *IEEE Transactions on Signal Processing*, 66(15), 4040–4053. <https://doi.org/10.1109/TSP.2018.2844203>
- Amari, S.-I. (1998). Natural gradient works efficiently in learning. *Neural Computation*, 10(2), 251–276. <https://doi.org/10.1162/089976698300017746>
- Amari, S.-I., Cichocki, A., & Yang, H. H. (1996). A new learning algorithm for blind signal separation. *Advances in Neural Information Processing Systems*, 8, 757–763.
- Bell, A. J., & Sejnowski, T. J. (1995). An information-maximization approach to blind separation and blind deconvolution. *Neural Computation*, 7(6), 1129–1159. <https://doi.org/10.1162/neco.1995.7.6.1129>
- Delorme, A., & Makeig, S. (2004). EEGLAB: An open source toolbox for analysis of single-trial EEG dynamics including independent component analysis. *Journal of Neuroscience Methods*, 134(1), 9–21. <https://doi.org/10.1016/j.jneumeth.2003.10.009>
- Delorme, A., Palmer, J., Onton, J., Oostenveld, R., & Makeig, S. (2012). Independent EEG sources are dipolar. *PLoS ONE*, 7(2), e30135. <https://doi.org/10.1371/journal.pone.0030135>
- Esmaeili, S. (2025). *Amica: Native python implementation of AMICA for MNE-Python and scientific EEG workflows*. <https://github.com/snesmaeili/amica>.
- Gramfort, A., Luessi, M., Larson, E., Engemann, D. A., Strohmeier, D., Brodbeck, C., Goj, R., Jas, M., Brooks, T., Parkkonen, L., & Hämäläinen, M. (2013). MEG and EEG data analysis with MNE-Python. *Frontiers in Neuroscience*, 7, 267. <https://doi.org/10.3389/fnins.2013.00267>
- Hannun, A., Digani, J., Katharopoulos, A., & Collobert, R. (2023). *MLX: Efficient and flexible machine learning on apple silicon*. <https://github.com/ml-explore/mlx>.
- Harris, C. R., Millman, K. J., Walt, S. J. van der, Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., & others. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362. <https://doi.org/10.1038/s41586-020-2649-2>
- Herforth, J. (2026). *Pyamica: PyTorch AMICA, adaptive mixture independent component analysis*. <https://github.com/DerAndereJohannes/pyamica>.
- Hyvärinen, A., & Oja, E. (2000). Independent component analysis: Algorithms and applications. *Neural Networks*, 13(4-5), 411–430. [https://doi.org/10.1016/S0893-6080\(00\)00026-5](https://doi.org/10.1016/S0893-6080(00)00026-5)
- Lee, T.-W., Girolami, M., & Sejnowski, T. J. (1999). Independent component analysis using an extended infomax algorithm for mixed subgaussian and supergaussian sources. *Neural Computation*, 11(2), 417–441. <https://doi.org/10.1162/089976699300016719>
- Palmer, J. A., Kreutz-Delgado, K., & Makeig, S. (2012). *AMICA: An adaptive*

- 162 *mixture of independent component analyzers with shared components.* Swartz
163 Center for Computational Neuroscience, University of California San Diego.
164 https://sccn.ucsd.edu/~jason/amica_a.pdf
- 165 Palmer, J. A., Kreutz-Delgado, K., Rao, B. D., & Makeig, S. (2008). Newton method for the
166 ICA mixture model. *2008 IEEE International Conference on Acoustics, Speech and Signal*
167 *Processing (ICASSP)*, 1805–1808. <https://doi.org/10.1109/ICASSP.2008.4517982>
- 168 Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin,
169 Z., Gimelshein, N., Antiga, L., & others. (2019). PyTorch: An imperative style, high-
170 performance deep learning library. *Advances in Neural Information Processing Systems*, 32,
171 8024–8035.
- 172 Pion-Tonachini, L., Kreutz-Delgado, K., & Makeig, S. (2019). ICLabel: An automated
173 electroencephalographic independent component classifier, dataset, and website.
174 *NeuroImage*, 198, 181–197. <https://doi.org/10.1016/j.neuroimage.2019.05.026>
- 175 Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D.,
176 Burovski, E., Peterson, P., Weckesser, W., Bright, J., & others. (2020). SciPy 1.0:
177 Fundamental algorithms for scientific computing in python. *Nature Methods*, 17(3),
178 261–272. <https://doi.org/10.1038/s41592-019-0686-2>